

Backside Power Delivery: Game Changer and Key Enabler of Advanced Logic Scaling and New STCO Opportunities

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Abstract—We report on devices built with the power delivery network (PDN) moved to the wafer's backside (BS) for lower IR drop and improved routing efficiency. This concept can be implemented by differentiated schemes, some of which can be combined for 3D stacked structures such as CFET, and with Buried Power Rail (BPR) connected via scaled nTSV to BSM1 used for a first experimental demonstration. Device fabrication starts with frontside (FS) processing, after which the wafers are flipped over, bonded to carrier wafers and continued for the BS flow. Robust extreme wafer thinning has been achieved with a SiGe-Etch Stop Layer (ESL) added early on the FS, without impacting device properties. Optimized post-BS anneal(s) are shown to be effective for V_T recovery (as some depassivation may occur after FS process), yielding improved mobility, DC performance, reliability and noise behavior. A scenario where the device's source (S) is directly contacted from the BS (BSC-S) offers higher cell's scalability potential, with stress, contact resistance, thermal (chip vs. transistor level, focus on hotspots) aspects assessed for various configurations. Its intrinsic S/D asymmetric access can also be explored to obtain low I_{OFF} for small bandgap Ge/SiGe FETs. Further exploration work shows clear benefits for ESD diodes with BS contacts and for clock distribution implemented with (partial) BS routing, paving the way to a truly functional BS and new STCO opportunities.

I. INTRODUCTION

With BSPDN, signal and power routing can be decoupled, allowing more room for their independent optimization (Fig.1) [1-4]. Key differentiators for various possible implementation options (examples in Fig.2) are the definition, within the cells, of the vertical connections between front and backside layers, their scalability potential and envisioned co-integration with advanced transistor architectures: from nanosheet (NS) FETs to 3D stacked FETs (e.g., CFET [5]). Thanks to its potential for enhanced power-performance benefits at system level (Fig.3), the concept is now embraced by all major industry players [6-8] as key to help further extend the scaling roadmap. Moreover, it represents a first step towards achieving a truly functional BS, part of a broader vision for improved system performance by leveraging, under the umbrella of STCO, the unique capabilities of logic, memory and 3D technologies into a potential future CMOS heterogeneous platform.

II. FIRST CONCEPT DEMONSTRATION

For an early concept demonstration we first built devices with nTSVs linking BSM1 to FSM1 pads [2], followed by a scheme with finFETs connected from both wafer sides via highly scaled nTSVs landing from the BS on BPR lines defined in the FS at varying lateral distances from the device's (outer)

fin [4,9] as illustrated by the TEM images in Fig.4, using the fabrication sequence shown in Fig.5. A critical module at the start of BS processing is the extreme wafer thinning, for which a thin SiGe-ESL (grown earlier in the FS flow as part of a SiGe/Si bilayer) was used to compensate for within-wafer non-uniformities of the Si substrate removal process; its presence confirmed to have no impact on device characteristics (Fig.6). Additionally, BS layers' lithography is challenged by bonding-induced wafer distortions, currently tackled using higher order scanner corrections for enabling tighter overlay control. TEMs from devices built on different thinned Si thicknesses (t_{Si}) are shown in Fig.7, with no t_{Si} impact on the DC characteristics seen for both N/PMOS after the various stages of fabrication (Fig.8). This change in t_{Si} (~210 vs. 110nm Si under BPR) does also not significantly affect the Kelvin and (BSM1-nTSV-BPR)-chain link resistance (R) values for the W-filled nTSVs used, as they are mostly due to contributions from the nTSV, its contact to BPR ($R_{contact}$) and the R of the $\frac{1}{2}$ BPR line per link in the chain. The negligible impact from extreme wafer thinning on device properties was further confirmed down to ~20nm-thick Si layer left under the STI oxide (Fig.9). It is also worth mentioning comparable matching results after FS vs. BS flows (Fig.10). $\Delta V_T < 0$ are, however, typically seen after BS processing, suggesting occurrence of some interface depassivation during this part of the flow, also since the V_T values can be mostly recovered by addition of a post-BS standard anneal at 420°C, 10% H_2/N_2 , 1atm (FGA). Post-BS anneal(s), as the last step(s) in the flow, can in fact be explored for device improvement. For example, a high pressure anneal (HPA) done at 100% H_2 , 20atm can lead to substantial mobility gains, especially for NMOS, as shown in Fig.11. This agrees with expected improvements from HPA on the interface status/passivation. At the same time, HPA is also thought to introduce excess hydrogen or charges in the bulk dielectric causing ΔV_T as seen in Figs.12 and 13. A subsequent FGA step can then help to effectively improve the bulk status, with minimized (or no) impact on the interface, provided its temperature is kept low enough ($\leq 350^\circ C$), as confirmed by the mobility, I_{ON} and D_{it} trends visible in Figs.11-14 for HPA $\rightarrow$ FGA (300 $\rightarrow$ 420°C). This post-BS anneal sequence was also verified to be beneficial in terms of LF-noise and BTI reliability as shown in Figs.15-17, providing further corroboration of the HPA benefits (especially seen for NMOS mobility, I_{ON} and PBTI) and the need for careful final FGA's temperature optimization to be able to preserve most of the HPA gains. For BPR-based devices, Fig.16 also indicates no BPR proximity effect on the BTI behavior as could be expected since the BPR is placed here at a ~50-60nm lower level than the fin's channel.

III. FURTHER SCALABILITY & EXPANSION POTENTIAL

Fig.18 illustrates two examples of standard cell layouts wherein devices are connected to BSPDN either via BPR or by contacting it directly to the transistors' sources through BSC-S [9,10]. The first approach's demonstration was described in the previous section, with VBPR vias connecting the BPR at the cell's bottom or top with the nearby transistor's source (through a M0 connection). The second scheme offers greater scaling potential as it avoids the need for a separate area inside the cell for BPR and VBPR placement, its feasibility also recently experimentally demonstrated [11]. Focusing on this last, ultra-compact case, minimizing the contact resistance for the BSC-S implemented from the wafer's BS will be key. Fig.19 shows that (including a distributed resistance in the contact model [9,12]) a larger contact surface area, namely for a more wrapped-around (WA) type of BSC contact, is expected to remain beneficial for various contact resistivities (ρ_{contact}).

Various BSC-S schemes were also evaluated for device level self-heating by 3D Monte Carlo simulations of bottom-cooled inverter cells (Fig.20) [9,13], assuming A14 HD design rules (DR) (42nm CPP, $L_{\text{gate}}=14\text{nm}$, 18nm M1 pitch, 4 stacked NS/FET). With no Si fins under the NS, part of the heat needs now to traverse the bottom-dielectric-isolation (BDI) causing a penalty vs. BPR reference. BDI's thermal optimization (e.g., $\downarrow$ thickness, $\text{SiO}_2 \rightarrow \text{SiN}$) helps; WA BSC-S (also beneficial for $\downarrow R_{\text{contact}}$) even more so, to the point of recovering baseline thermal performance, as it allows heat to escape sideways from epi areas. Self-heating is inherently a multi-scale interplay of thermal transport at device and chip/system levels, such that, depending on packaging and cooling scenarios, most heat can end up being removed via the cell's topside (Fig.21, scheme). Chip level heat diffusion simulations [14] with detailed account for the distinct thermal conductivity sequence of different transistor/cell topologies yield near identical chip level temperatures (Fig.21, left) since the cells are only a minor fraction of the total junction-to-sink thermal resistance. Even so, BSC-S can still alleviate hotspots that arise locally relative to the chip background in circuit regions with much higher activity factors, e.g., the buffer cells in Fig.21, right.

Stress-wise, evaluating its evolution throughout the flow (Fig.22) indicates limited impact from BS processing [9,12]. During BSC-S etch, widening the opening area and partially etching into the source-epi (S-epi) is seen in Fig.23 to not detrimentally impact the top (as seen from the BS) channel's stress. On the contrary, some (small) stress enhancement seems to occur linked to the device/stacks configuration in place. This allows considering some interesting options for device engineering namely, as depicted in Fig.24, BSC-S etch into S-epi (potentially V-shape like inside the epi for increased contact surface area) followed by the growth of a thin, highly doped epi layer for lower R_{contact} . With the BSC-S scheme, source/drain (S/D) access is also intrinsically asymmetric which could be used to obtain, in a simpler way, devices with lowly doped drain and highly doped source regions as described in Fig.25, thus enabling, e.g., low I_{OFF} for small bandgap Ge/SiGe FETs [15].

When performing BSPDN scheme selection, co-integration with novel transistor architectures ahead in the roadmap (e.g.,

3D stacked structures like CFET) also needs to be anticipated. For CFET, a combination of several approaches first evaluated in NSFETs will likely be implemented as schematically illustrated in Fig.26, namely with BSC-S used for the bottom device and one of the many possible flavors of TSVM used for the top device. A first performance assessment (Fig.27) for 5T vs. 4T height HD cells using NSFETs and BPR (assuming A14 DR) or CFET (A5 DR with 39nm CPP, $L_{\text{gate}}=12\text{nm}$, 16nm M1 pitch, 2 stacked NS/FET; $\neq$ NS widths; simple version of BSC-S for the bottom device), respectively, indicates feasibility of energy and power gains for the more compact cells, with several knobs available for further speed exploration.

BSPDN, besides being a game changer for on-chip power distribution, also paves the way towards a truly functional BS roadmap, as depicted in Fig.28, with the potential for enhanced system performance. An early insight into the benefits of expanding the concept towards other functions, namely by addition after BS processing of specific devices (e.g., for I/O and ESD [16], MIM capacitors [17]) or structural interconnects (e.g., for clock distribution [18]) is provided in Figs.29 and 30. Partly using BS routing for the clock's Htree portion of a HP CPU (at A14 DR) results in $\sim 20\text{ps}$ latency from the Htree as seen in Fig.29, whereas for a conventional, fully FS routed case, that value rises to $\sim 50\text{ps}$ resulting, on average, in a $\sim 30\text{ps}$ latency increase for the whole chip (Fig.29b). Moving larger amounts of the clock signal to the BS is expected to lead to further latency reduction, with careful optimization needed to avoid running into BS (power and signal) routing congestion issues. Using the wafer's BS also allows for more compact ESD vertical diodes with BS contacts, as shown in Fig.30, which further exhibit improved current uniformity values as compared to FS only versions. Improved latch-up immunity can also be obtained thanks to lower parasitic bipolar current gain (β) values as seen both experimentally from parasitic npn bipolar transistors ($\downarrow \beta$ for $\downarrow t_{\text{tsi}}$, further reduced with BS contacts on p+/p-well areas - Fig.30b) and also confirmed from simulations (Fig.30c) yielding $\sim 45\%$ lower β for an optimized layout of the npn configurations with BS base terminals.

IV. CONCLUSIONS

BSPDN is an innovative and game changer approach for on-chip power delivery, key for supporting scaling roadmaps. With no detrimental N/PMOS impact from BS processing assessed by early experimental demonstrations, this concept can also allow new device engineering opportunities and expand towards a more optimum use of both wafer sides (e.g., adding functions/devices on BS) for enhanced system performance.

REFERENCES

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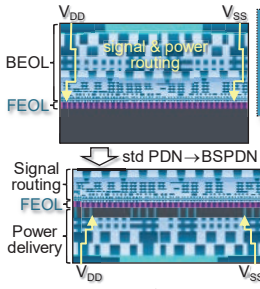


Fig.1 – BSPDN is a new game changer concept for on-chip power distribution. Signal and power wiring can be decoupled by using both wafer sides for routing, with the latter moved to the wafer's backside.

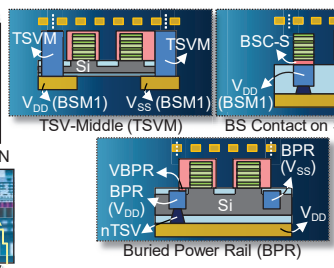


Fig.2 – Schematics illustrating various differentiated BSPDN implementation options. Connection to the BS power is done via a BPR or TSV-middle scheme or by contacting the transistor's source directly from the wafer's BS (BSC-S).

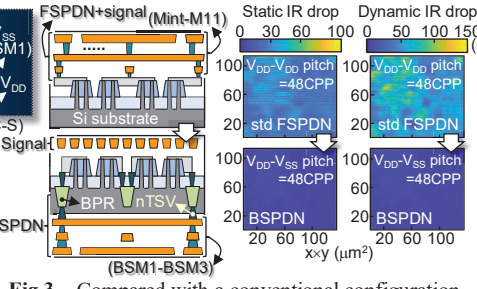


Fig.3 – Compared with a conventional configuration for signal and PDN routing (FSPDN), BSPDN using nTSV landing on BPR and extreme thinned wafers offers a clear advantage, for both dynamic and static modes, with tighter distributions & smaller IR drop values computed from on-chip IR drop heat maps for a low power 64-bit CPU ($V_{DD}=0.7V$, A14 DR).

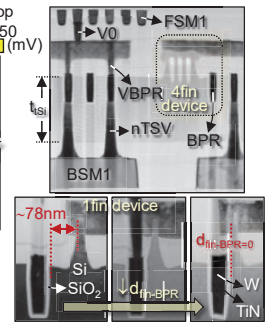


Fig.4 – TEMs of finFETs connected using both wafer sides via W-filled BPR lines, placed at varying lateral distances from the (outer) fin.

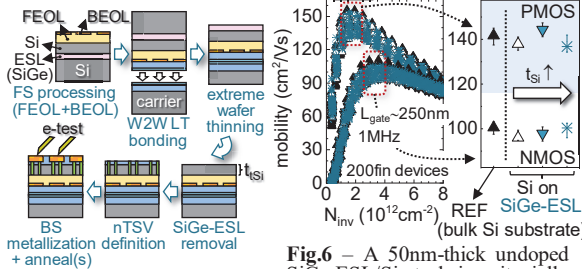


Fig.5 – Key fabrication blocks to build devices with BSPDN-BPR. A SiGe-ESL enables a more robust wafer thinning process. BS layers' exposures, due to bonding-induced wafer distortions, use higher order scanner corrections for tighter overlay control.

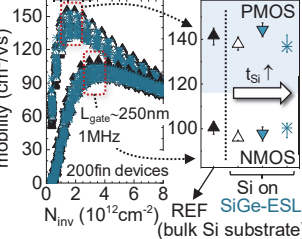


Fig.6 – A 50nm-thick undoped SiGe-ESL/Si stack is epitaxially grown early in the FS part of the flow. Its good epi quality is confirmed by physical analyses and by devices exhibiting similar characteristics (e.g., mobility as shown here) vs. reference FETs built on bulk Si substrates.

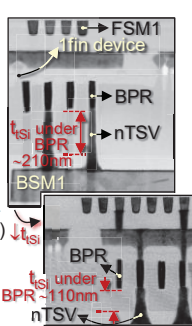


Fig.7 – TEM images of devices built on wafers with ~210 or ~110nm thinned Si layer (t_{Si}) under the BPR lines.

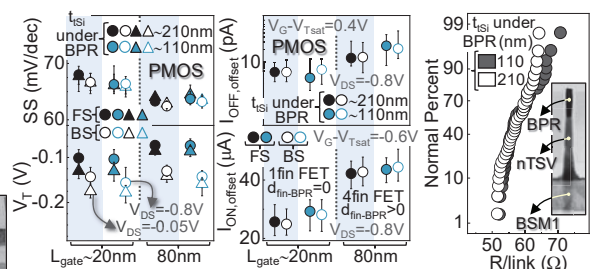


Fig.8 – N/PMOS built on wafers with different thinned Si thicknesses (as in Fig.7) exhibit similar device characteristics after both FS and BS processing, and for various L_{gate} , as shown here by several FOMs for PMOS. These t_{Si} s also correspond to comparable (BSM1-nTSV-BPR)-chain link resistances, mostly due to the contributions from the nTSV, its contacting to BPR (nTSV-BPR contact resistance), and the $\frac{1}{2}$ BPR line in the link.

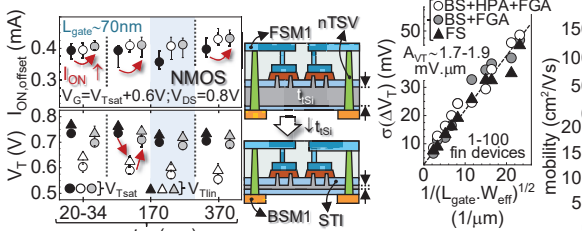


Fig.9 – No significant impact on device characteristics (comparing data after FS vs. BS (+anneal) processing) is also seen for extreme thinned Si substrates. Shown here results down to a ~20nm-thick Si layer left under the STI oxide for NMOS built on wafers with nTSVs landing directly on FSM1 as illustrated by the simple schematics on the right.

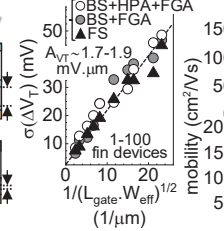


Fig.10 – No impact from BS processing is seen on the matching performance of finFET devices. A_{VT} values are in line with those obtained from devices built with a standard configuration using FS only connections.

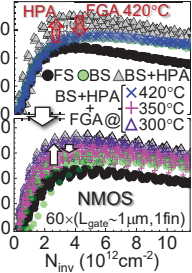


Fig.11 – Higher mobility values are obtained after post-BS anneal, especially HPA and for NMOS. Extra FGA needs to be done at a lower temperature ($T_{FGA} < 350^\circ\text{C}$) in order to preserve HPA gains.

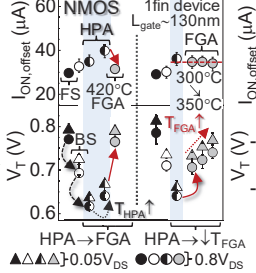


Fig.12 – Agreeing with the results in Fig.11, applying a final FGA with lower T_{FGA} (instead of the 420°C standard FGA) allows keeping most of the I_{ON} gains from a post-BS HPA, and still be effective in V_T recovery (from BS shifts).

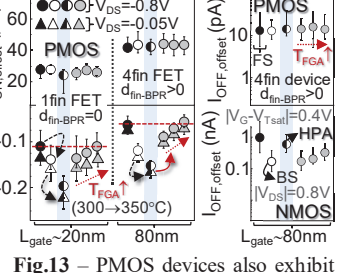


Fig.13 – PMOS devices also exhibit progressive V_T recovery with T_{FGA} after BS process and HPA. At 350°C T_{FGA} , they show V_T values similar to those after FS fabrication. No I_{ON} (at V_T overdrive) impact from T_{FGA} is seen in the $300\text{--}350^\circ\text{C}$ range. For both P/NMOS there is also no I_{OFF} degradation from BS and post-BS anneal(s) processing (on the right).

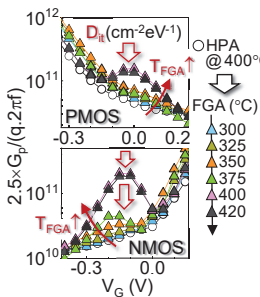


Fig.14 – For both N/PMOS, HPA benefits are confirmed by low D_{It} values. They are kept with a subsequent FGA at $\leq 350^\circ\text{C}$, slightly increasing at 375°C for NMOS, more strongly (N/PMOS) $\geq 400^\circ\text{C}$.

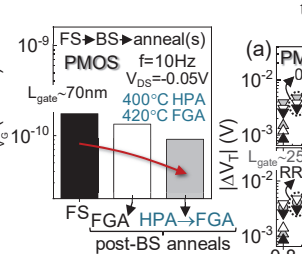


Fig.15 – Addition of an HPA treatment just after BS processing, and prior to a final FGA step, is also seen to lead to overall improved LF-noise values, with the devices built exhibiting typical $1/f$ noise behavior.

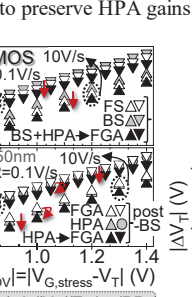


Fig.16 – Constant ramped voltage stress BTI measurements were done at 25°C for three ramp-rates ($RR = 0.1, 1, 10V/s$). Agreeing with the noise results, PMOS exhibit reduced $|\Delta V_T|$ at a given overdrive voltage ($|V_{ov}|$) for HPA+FGA after BS process (a). Also, no BTI impact from BPR proximity is seen (b).

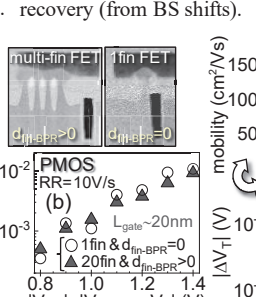


Fig.17 – For NMOS, adding a post-BS HPA is seen to be highly beneficial for BTI behavior (smaller $|\Delta V_T|$ and A (calculated from all RR) for similar γ, n), with an extra, higher T_{FGA} (420°C) step also reducing, as for mobility, the gains.

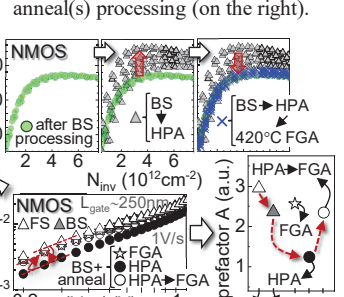


Fig.18 – Mobility vs. N_{inv} plot for NMOS devices, showing improved performance after HPA and FGA.

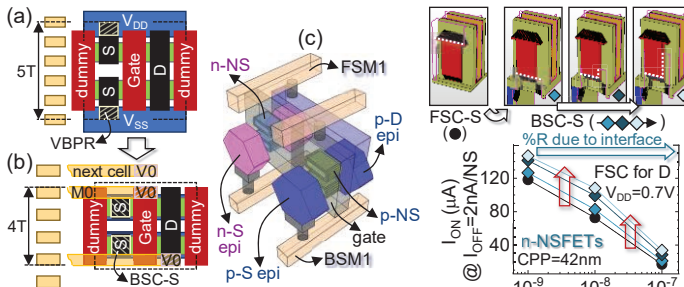


Fig.18 – Examples of standard cell layouts where devices are connected to BSR either via BPR (a) as in Fig.4, or by directly contacting their sources from the BS via BSC-S (b,c). The latter enables further cell's scalability (e.g., 5T→4T cell height in (a,b)), avoiding BPR & VBPR vias placement on the floorplan.

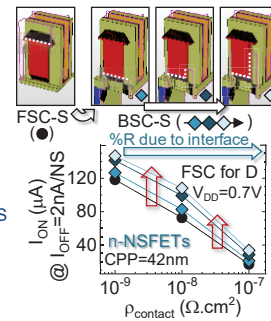


Fig.19 – As for devices connected only from the FS, having a larger contact surface area for BSC-S is also predicted by TCAD to be beneficial for reducing R_{contact} and thus enable higher performance.

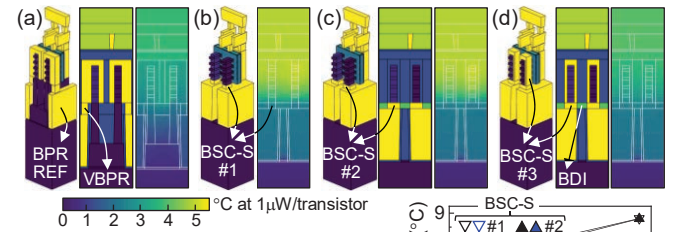


Fig.20 – Self-heating (relative to cell bottom) assessment of NSFETs with various BSC-S configurations vs. a reference BPR case. 3D geometries and corresponding thermal cross-section maps (with cut across the source terminal) shown in (a-d). A comparison of the devices' local temperature rise values (n/p channels' heating average) highlights the benefits also from a thermal perspective of implementing a more WA type of BSC-S and of using a thinner BDI with a more thermally suitable material such as SiN vs. SiO_2 .

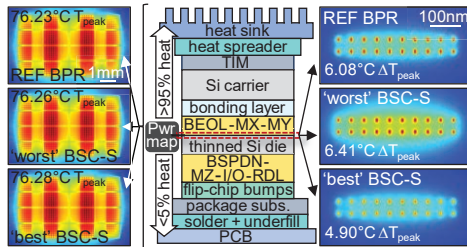


Fig.21 – At chip level (temperature maps on the left for a SOC of 4x4 mempool clusters with L2 caches, 8x5mm² Si die, A14 HD NSFET design rules) similar results are computed for the various transistor configurations from Fig.20 as nearly all heat is removed upwards in the depicted multi-layer stack schematics. Zooming-in for hotspots (on the right-hand side), as the heat can in this case spread sideways and flow in both directions, that leads to a different outcome with the best BSC-S scenario in Fig.20 shown then to be beneficial.

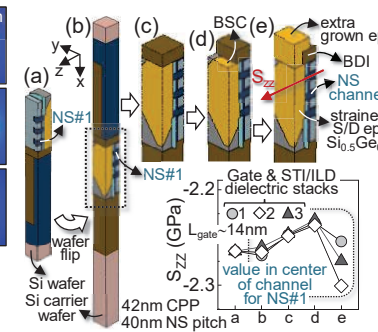


Fig.22 – Simulated channel stress evolution throughout the flow after key fabrication stages (PMOS): (a) FS processing; (b) wafer flip and bonding; (c) substrate thinning down to BDI; (d) BSC etch; (e) extra $\text{Si}_{0.5}\text{Ge}_{0.5}$ epi layer growth on open contact area. Overall, stress impact is limited.

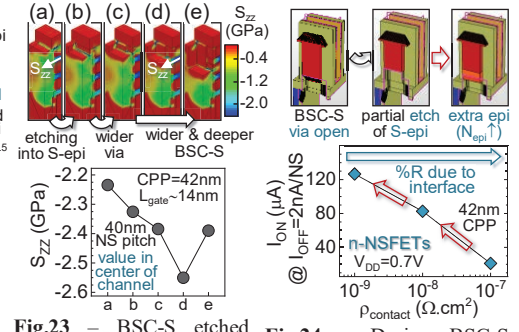


Fig.23 – BSC-S etched wider and deeper into the S-epi does not seem to detrimentally impact the top (as seen from the BS) NS's channel strain. On the contrary, it appears to lead to some (small) interesting simulated stress increase.

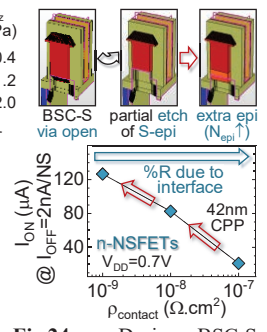


Fig.24 – During BSC-S definition, partially etching into the S-epi can widen the options for R_{contact} reduction, namely (as shown in the schematics) with the growth of an extra low temperature, highly doped epi layer.

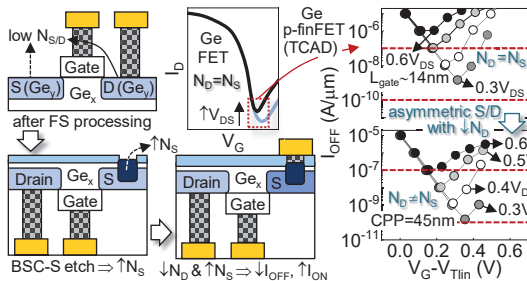


Fig.25 – Devices with BSC-S are intrinsically asymmetric for S/D access as the drain terminal remains contacted from the FS. This can be explored to build transistors with lowly doped D (defined during the FS flow) and highly doped S (after BSC-S etch) areas as illustrated here for a generic device architecture. An interesting option for high mobility, low bandgap (e.g., Ge) FETs that, despite high drivability potential, typically also suffer from higher off-state leakage (I_{OFF}), and for which this scheme can help both I_{ON} and I_{OFF} (e.g., TCAD data, right).

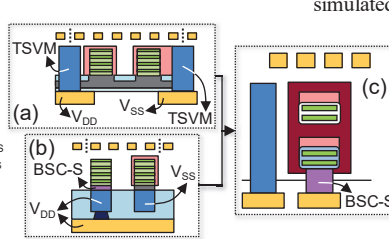


Fig.26 – BSPDN for 3D stacked CMOS like the so-called CFET, where NMOS and PMOS are folded on top of each other, can be envisioned to consist in a combination of several of the integration options first considered in the context of NSFETs such as the BSC-S approach for the bottom device (b→c) and one of many flavors of the TSMV concept for the top FET (a→c).

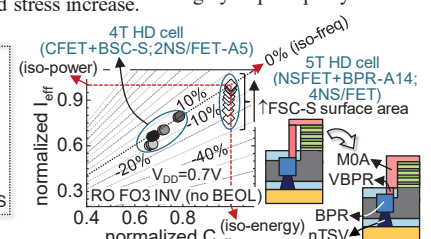


Fig.27 – A first performance assessment of various NSFET (with BPR) vs. CFET (with a simple version of BSC-S assumed for the bottom device) configurations (≠ NS number and width per device) for 5T vs. 4T height HD cells, respectively. The results show clear power and energy gains seem feasible for scaled CFET cells, with several knobs further possible to explore for speed.

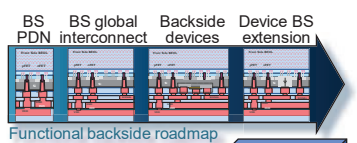


Fig.28 – BSPDN has the potential to further expand with migration of system functions to the BS. Addition of specific devices (e.g., I/O, ESD, MIM capacitors) or structural interconnects (e.g., clock distribution) can pave the way to a truly functional BS for enhanced system performance.

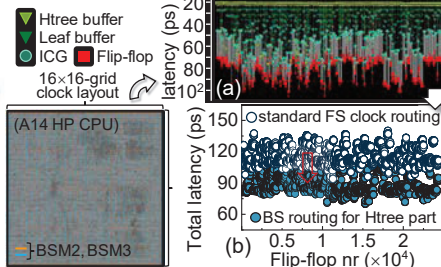


Fig.29 – Early exploration of introducing BS routing for the clock's Htree part (global routing with 16x16 grid) of a HP CPU at A14 design rules (example of clock latencies distribution in (a)). Benefits of even only partial clock move to the BS highlighted in (b).

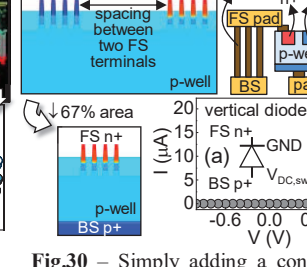


Fig.30 – Simply adding a contact on the wafer's backside during BS processing can enable through-wafer devices such as vertical ESD diodes (~3x more compact), as demonstrated experimentally in (a). Measured (b) and simulated (c) results also show latch-up prevention benefits from lower current gains (β) for npn bipolar transistors with BS p-well contacts.